

D-18/2113
DATA STRUCTURES-2302T
(Semester-III)

Time : Three Hours]

[Maximum Marks : 50

Note : Attempt any *three* questions from Section-A and any *three* questions from Section-B. Section-C is compulsory.

SECTION-A

- I. What is Stack? Write the algorithm for PUSH and POP operation on stack in sequential representation. 5
- II. Write the algorithm for Bubble sort and discuss its complexities. 5
- III. What is Linked list? Write the algorithm to insert a node in a sorted singly linked list. 5
- IV. What is Hashing? What do you understand by collision in it? Discuss the collision handling using chaining. 5
- V. (a) What is Binary search? Write the recursive algorithm for binary search.
(b) What do you understand by Circular queue? Illustrate its advantages over simple queue. 5

SECTION-B

- ~~VI.~~ How does Dijkstra's algorithm work, and what is its application in finding shortest paths in graphs? 5
- VII. What is a Binary Search Tree? Write the algorithm to insert a node in a Binary Search Tree. 5
- VIII. What is the minimum spanning tree (MST) problem, and how are algorithms like Prim's used to find MSTs in graphs? 5
- IX. (a) Explain the difference between a primary index and a secondary index in indexed file organization.
- (b) How are rotations (left rotation and right rotation) used to maintain balance in an AVL tree? 5
- X. How do pre-order, in-order, and post-order differ in terms of node visitation order? Discuss. 5

SECTION-C
(Compulsory Question)

- XI. (a) What is a balanced tree, and why are balanced trees like AVL trees important in data structures? 2
- (b) What are the primary goals of file organization? 2
- (c) What is the balance factor of a node in an AVL tree, and how is it calculated? 2
- (d) What is the role of data structures in designing efficient algorithms for sorting and searching? 2

- (e) Write the algorithm for linear search. 2
 - (f) Discuss in brief the applications of queue data structure. 2
 - (g) What do you understand by a binary tree? 2
 - (h) Discuss the concept of priority queues and their use cases. 2
 - (i) How can a stack be used to reverse a sequence of elements? 2
 - (j) Discuss advantages and disadvantages of linked lists compared to arrays in terms of memory usage and operations like insertion and deletion. 2
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SECTION—A

1. What do you mean by an array? Write algorithm to insert and delete elements from an array. 5

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[P. T. O.]

2. What are Stacks? Write procedures for PUSH and POP operation. 5
3. Explain the advantages and disadvantages of doubly linked list over singly linked list. 5
4. Compare linear and binary search with suitable examples. 5
5. Write algorithm for quick sort and discuss same with the help of examples. 5

SECTION—B

6. What is traversing of binary tree? Draw a tree for Preorder and Inorder traversing. 5
7. Discuss insertion and deletion in binary tree. 5
8. Explain Dijkstra's algorithm for shortest distance with the help of suitable examples. 5
9. What is Minimum spanning tree (MST)? How prim's algorithm used to find MST's in graph. 5

10. Discuss different file organization techniques. 5

SECTION—C

11. Answer the following questions : $10 \times 2 = 20$

- (i) Compare stack and queue.
- (ii) Write the algorithm for linear ^{search} algorithm.
- (iii) What is Priority queue?
- (iv) Define the Graph.
- (v) Compare static memory allocation and dynamic memory allocation.
- (vi) Time and space complexity.
- (vii) Hash function.
- (viii) AVL tree.
- (ix) Discuss chaining in Hashing.
- (x) Define the Heap.

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DATA STRUCTURES (CPE-203)

3rd Semester

Time: 03 Hours

Max. Marks: 50

Instructions to Candidates:

- 1) Attempt any **three** questions from Section A
- 2) Attempt any **three** questions from Section B.
- 3) Section C is compulsory.

Section- A (3 x 5 = 15)

- Q 1) What is Stack? Why it is known as LIFO? Write algorithms for PUSH and POP.
- Q 2) What is dynamic memory allocation? How linked list helps in dynamic memory allocation.
- Q 3) Explain different techniques of hashing? Also discuss collision resolution techniques.
- Q 4) How does a binary search algorithm work? Demonstrate with example.
- Q 5) Write an algorithm that inserts an element ELEMENT into kth position of linear array.

Section –B (3 x 5 = 15)

- Q6) Explain Inorder, Preorder and Postorder Traversal operation on Binary tree with example.
- Q 7) What are the different representations of graphs in memory?
- Q 8) Explain the process of depth first traversal of a graph by taking a suitable example.
- Q9) Discuss different file organization techniques.
- Q 10) Write down the algorithm for insertion and deletion in a binary search tree.

Section –C

Section –C

Q 11)

(10 x 2 =20)

- a) Compare Static Memory Allocation and Dynamic Memory Allocation.
- b) Define a graph.
- c) Which data structure is used to perform recursion and why?
- d) Give some applications of stack.
- e) What is the difference between data, data type and data structure?

questions each from Sections A and B carrying

SECTION-A

- f) What is garbage collection?
- g) Define multi linked structures.
- h) What are records?
- i) When do we use priority queues?
- j) How can we calculate the address of any element of a 2D array?

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DATA STRUCTURES-203

(Semester-III)

Time : Three Hours]

[Maximum Marks : 50

Note : Section C is compulsory. Attempt any *six* questions by selecting *three* questions each from Section A and B.

SECTION-A

I. How arrays are stored in memory? Explain column major representation of an array. (5)

II. Explain conversion from infix to postfix representation with the help of stack use. (5)

✓ III. Explain the linked representation of a circular queue and operations to be performed on it with the help of suitable example. (5)

✓ IV. What is a Linked List? Explain the algorithm for insertion of a node in a Linked List. (5)

✓ V. What is Selection sort? Explain with the help of suitable example. (5)

SECTION-B

- ✓ VI. What is a graph? How graphs are represented in computer memory? Show with the help of suitable example. (5)
- ✓ VII. Write the algorithm for post-order tree traversal. Also show the steps of this algorithm on a set of numbers to show an example. (5)
- VIII. The Keys 12, 18, 13, 2, 3, 23, 5 and 15 are inserted into an initially empty hash table of length 10 using open addressing with hash function $h(k) = k \bmod 10$ and linear probing. What is the resultant hash table? (5)
- IX. Explain Dijkstra's Algorithm for shortest distance with the help of suitable example. (5)
- ✗ X. What is minimum spanning tree? Explain the Kruskal's algorithm for finding minimum spanning tree. (5)

SECTION-C
(Compulsory Question)

XI. Answer the following in brief :

✓ (a) Linear search vs. Binary search.

✓ (b) Abstract Data Type.

✓ (c) Hash function.

✓ (d) Data structures for searching.

✓ (e) Linked representation of queue.

✓ (f) Binary Search Tree.

✓ (g) Depth First Search.

✓ (h) Heap.

✓ (i) AVL Tree.

✓ (j) Operations on Graph.

(2×10=20)